

JanGyan — Learn Without Limits

Category : Cyber Safety & Digital Security

Module 7 - Device Security

Module Overview

This handbook is your complete, beginner-to-advanced guide to **Device Security** — covering Android, Windows, software updates, antivirus protection, and app permissions. Your phone and computer are the front door to almost everything digital in your life. This handbook shows you how to keep that front door properly locked.

By the end of this module, you will not need to search anywhere else to understand how to secure the devices you use every day.

Learning Objectives

After completing this module, you will be able to:

- Understand how Android and Windows protect you by default, and where their limits are
- Explain why software updates are one of the most important security habits
- Choose and use antivirus protection appropriately
- Review and manage app permissions confidently
- Recognise the risks of sideloading apps and using unsupported software
- Apply a practical device-security routine to your phone and computer

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Why This Topic Matters

Google's built-in Android protection, Play Protect, now scans over **350 billion apps daily** and blocked **266 million risky sideload attempts** in 2025 alone — yet malicious apps from outside the Play Store nearly doubled that same year, rising from 13 million detected in 2024 to **27 million in 2025**.

> *Reality Check: Windows 10 reached the end of standard support on 14 October 2025. Free Extended Security Updates for individual users run only until 13 October 2026 — meaning any device still running Windows 10 without enrolling in that program will soon stop receiving critical security patches entirely.*

Every device-security decision — installing an update, granting a permission, choosing an app source — either closes a door to attackers or leaves it open. Most of these decisions take less than a minute; the protection they provide lasts far longer.

Introduction

Device security covers everything that keeps your phone, laptop, or computer resistant to malware, unauthorised access, and data theft. It rests on five pillars covered in this module:

- **Operating System (Android/Windows):** The foundation layer with its own built-in protections

- **Updates:** Regular patches that close known security gaps
- **Antivirus:** Software that detects and blocks malicious programs
- **App Permissions:** Controls over what each app can access on your device
- **Safe Habits:** Choosing trusted sources and avoiding risky shortcuts like sideloading

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Fundamental Concepts

Patch vs Feature Update

Type	Purpose
Security Patch	Fixes a specific vulnerability that could be exploited by attackers
Feature Update	Adds new functionality or design changes, not primarily security-focused

The Permission Model

Modern Android and app ecosystems use **runtime permissions** — an app must ask for access (camera, location, contacts, microphone) at the moment it needs it, and you can grant, deny, or later revoke that access individually.

Built-in vs Third-Party Antivirus

Type	Examples	Note
Built-in Protection	Google Play Protect (Android), Microsoft Defender (Windows)	Free, automatically active, regularly updated
Third-Party Antivirus	Independent security software	Can add extra features, but should not be assumed automatically superior

A strong antivirus is a safety net, not a substitute for safe habits — updates and cautious permission management matter just as much.

Complete Detailed Explanation

Part A — Android Security

- **Google Play Protect:** Android's built-in malware scanner, running continuously in the background, scanning installed and newly downloaded apps against Google's threat intelligence.
- **Sideloading Risk:** Installing APK files from outside the Play Store bypasses Play Store's review process; in 2025, sideloaded apps were the dominant source of new Android malware detected.
- **Anti-Social-Engineering Protection:** Android now prevents users from disabling Play Protect while on an active phone call — a direct response to scams that talk victims into turning off protection before installing malicious software.

- **App Permissions:** Android's runtime permission system lets you grant or deny access to camera, location, contacts, and more, individually per app, and revoke them later at any time.

Part B — Windows Security

- **Windows Update:** Microsoft's mechanism for delivering security patches, feature updates, and driver fixes — should generally be left on automatic.
- **Microsoft Defender:** Windows' built-in antivirus and firewall solution, enabled by default and continuously updated.
- **Windows 10 End of Support:** Windows 10 reached the end of standard support on 14 October 2025. Devices not enrolled in Extended Security Updates (ESU) no longer receive routine security patches, leaving newly discovered vulnerabilities permanently unpatched.
- **Extended Security Updates (ESU):** A bridge program providing critical/important security patches only (no new features or general support) for enrolled Windows 10 devices, running through 13 October 2026 for individual users.
- **Windows 11 Security Features:** Requires hardware-based protections like TPM (Trusted Platform Module) and Secure Boot, raising the baseline security level compared to older Windows versions.

Part C — Software Updates

Why updates matter: Nearly every major cyberattack exploits a vulnerability that a patch already exists for — attackers specifically target devices that have delayed installing available updates.

Best practice: Enable automatic updates wherever possible, for the operating system, browser, and individual apps alike, rather than manually postponing them.

Part D — Antivirus Protection

What it does: Scans files and apps for known malicious signatures and suspicious behaviour, blocking or quarantining threats before they cause harm.

What it doesn't do: Antivirus cannot stop you from voluntarily sharing an OTP, password, or personal data with a scammer — it protects against malicious software, not social engineering.

Part E — App Permissions

Principle of least privilege: An app should only be granted the permissions it genuinely needs for its core function.

Red flags: A flashlight, calculator, or wallpaper app requesting access to contacts, SMS, or location has no legitimate need for that data — deny such requests or avoid the app.

Regular review: Periodically check your phone's app permission settings and revoke access that's no longer necessary or was granted by mistake.

Important Terminologies

Term	Simple Meaning
Patch	A software update that fixes a specific security vulnerability
Sideloading	Installing an app from outside the official app store
Runtime Permission	An app's request for access to a specific feature (camera, location) at the time it's needed
Google Play Protect	Android's built-in malware-scanning system
Microsoft Defender	Windows' built-in antivirus and firewall protection
TPM (Trusted Platform Module)	A hardware security chip that supports encryption and secure boot processes
Secure Boot	A feature ensuring only trusted software loads when a device starts up
Extended Security Updates (ESU)	A paid/limited program providing continued security patches after a product's official support ends
Principle of Least Privilege	Granting only the minimum access needed for a task, nothing more

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Step-by-Step Understanding

How to Check and Update Your Devices

1. **Android:** Settings → System → Software Update → Check for updates
2. **Windows:** Settings → Windows Update → Check for updates
3. Enable "Automatic Updates" wherever the option is available
4. Restart your device promptly when an update requires it, rather than postponing indefinitely

How to Audit Your App Permissions

1. Open Settings → Apps → Permissions (Android) or Privacy & Security (Windows)
2. Review which apps have access to camera, microphone, location, and contacts
3. Revoke access for any app that doesn't need it for its core function
4. Uninstall apps you no longer use rather than leaving them with standing permissions
5. Repeat this review every few months

Real-Life Indian Examples

(Widely reported public patterns, described for educational awareness.)

- **Fake APK Scams via WhatsApp:** Fraudulent "courier tracking," "electricity bill," or "government form" APK files shared through WhatsApp have repeatedly been flagged by Indian cybersecurity teams as a common trojan-delivery method, especially targeting less tech-familiar users.
- **Seasonal Malicious App Surges:** Security researchers have repeatedly observed spikes in fake shopping, loan, and festival-discount apps around major Indian shopping seasons, many later found to request excessive permissions or contain hidden malware.
- **Outdated Device Exploitation:** Devices running unsupported or outdated operating systems have been common targets in reported malware campaigns, since known vulnerabilities on them remain permanently unpatched.

Practical Scenarios

Scenario 1: A "torch/flashlight" app requests permission to read your contacts and SMS messages.
Deny the request — a flashlight app has no legitimate need for this data. Consider using your phone's built-in flashlight feature instead.

Scenario 2: You keep postponing a Windows security update notification because you're busy.
Schedule the update for a convenient time rather than skipping it repeatedly — unpatched vulnerabilities are actively targeted by attackers.

Scenario 3: A friend sends you an APK file directly on WhatsApp instead of a Play Store link.
Avoid installing it. Search for the app directly on the Play Store instead, where it goes through Google's review process.

Scenario 4: Your laptop is still running Windows 10 and you're unsure if it's still protected.
Check whether you're enrolled in Extended Security Updates; if not, plan to upgrade to a supported Windows version before the ESU program ends.

Government Rules / Guidelines

- **CERT-In:** Issues advisories on device security, malicious apps, and patch management, and operates the Cyber Swachhta Kendra initiative offering free malware-removal tools to citizens, with coverage reported across roughly 98% of India's digital population.
- **CISG-2025-02 Guidelines (2025):** CERT-In's mandate requiring regular third-party cybersecurity audits for many digital service providers, indirectly strengthening the security of apps and platforms citizens use daily.

Laws & Policies

Law / Provision	What It Covers
IT Act, 2000 — Section 43	Civil liability for unauthorised access, damage, or data loss involving computer systems and devices
IT Act, 2000 — Section 66	Criminal liability for hacking and computer-related offences, including those exploiting unpatched or compromised devices
CERT-In Directions (2022, ongoing)	Require reporting of significant cyber incidents within 6 hours, encouraging organisations to maintain updated, well-monitored systems

Important Statistics

Metric	Figure
Apps scanned daily by Google Play Protect	Over 350 billion
Risky sideload attempts blocked by Play Protect (2025)	266 million
New malicious apps detected outside Play Store (2025)	27 million, up from 13 million in 2024
Policy-violating apps blocked from Google Play (2025)	1.75 million
Developer accounts banned by Google (2025)	Over 80,000
Apps stopped from over-accessing user data (2025)	Over 255,000
Windows 10 end of standard support	14 October 2025
Windows 10 consumer Extended Security Updates end	13 October 2026
Cyber Swachhta Kendra digital population coverage	About 98%

(Figures are approximate, drawn from Google, Microsoft, and CERT-In/PIB publications through 2025–2026.)

Best Practices

- Enable automatic updates for your operating system, browser, and apps
- Install apps only from official stores (Google Play, verified sources) — avoid sideloading APKs
- Review app permissions regularly and revoke anything unnecessary
- Keep built-in protections (Play Protect, Microsoft Defender) enabled at all times
- Plan operating system upgrades before official support ends
- Restart your device promptly when updates require it

Common Mistakes

- Repeatedly postponing operating system or app updates
- Installing APK files sent directly via WhatsApp or other messaging apps
- Granting every permission an app requests without reviewing them
- Disabling built-in antivirus/protection features to install unfamiliar software
- Continuing to use an unsupported operating system without any additional protection
- Assuming antivirus alone makes risky habits (like sharing OTPs) safe

Myths vs Facts

Myth	Fact
"If my phone/computer works fine, I don't need to update it"	Unpatched vulnerabilities remain exploitable even if a device seems to work normally
"Antivirus software alone keeps me completely safe"	Antivirus doesn't protect against social engineering, like sharing an OTP with a scammer
"Apps outside the Play Store are just as safe if they look legitimate"	Sideloaded apps bypass Play Store review and were the leading source of new Android malware in 2025
"Windows 10 will keep receiving free security updates indefinitely"	Free consumer Extended Security Updates end on 13 October 2026
"A flashlight or simple utility app needs broad permissions to work well"	Simple utility apps rarely need access to contacts, SMS, or location

Do's and Don'ts

Do	Don't
Enable automatic updates on all devices	Don't repeatedly postpone security update notifications
Install apps only from official app stores	Don't sideload APK files from messaging apps or unofficial sites
Review app permissions regularly	Don't grant every permission an app requests by default
Keep built-in antivirus/protection enabled	Don't disable security features to install unfamiliar apps
Plan upgrades before OS support ends	Don't continue using an unsupported OS without any mitigation plan

Safety Tips

For Android Devices
- Keep Google Play Protect enabled at all times
- Avoid installing APKs from WhatsApp, SMS, or unofficial websites
- Review app permissions every few months and remove unused apps

For Windows Devices
- Keep Windows Update and Microsoft Defender enabled
- Check whether your device is eligible for Windows 11 or needs ESU enrollment
- Avoid downloading software from unofficial or pirated sources

For Families
- Set up devices for children and seniors with automatic updates enabled by default
- Review installed apps and permissions together periodically
- Explain why "just this once" skipping an update isn't worth the risk

For Everyone
- Treat every unexpected APK/file request with suspicion
- Restart devices when prompted after updates, rather than delaying indefinitely
- Keep only apps you actively use installed on your device

Practical Checklist

- ☐ Enabled automatic updates on phone and computer
- ☐ Confirmed Google Play Protect / Microsoft Defender is active
- ☐ Reviewed and revoked unnecessary app permissions
- ☐ Removed unused or unfamiliar apps
- ☐ Checked whether my Windows device needs ESU enrollment or an upgrade
- ☐ Avoided installing any APKs from outside the Play Store
- ☐ Shared these habits with family members' devices too

Quick Revision Notes

- Device security rests on: OS protections, updates, antivirus, app permissions, and safe habits
- Updates patch known vulnerabilities — postponing them leaves devices exposed
- Google Play Protect and Microsoft Defender are strong, free, built-in protections
- Sideloading APKs bypasses app store review and is the leading Android malware source
- Windows 10 free security updates end 13 October 2026 — plan upgrades ahead of this
- Review app permissions regularly using the principle of least privilege

Key Takeaways

1. An update you postpone today is a vulnerability an attacker can use tomorrow
2. Built-in protections (Play Protect, Microsoft Defender) are strong — keep them on
3. Sideloading apps trades convenience for a real, measurable increase in risk
4. App permissions should always match what an app genuinely needs to function
5. Antivirus protects your device from malware — it can't protect you from voluntarily sharing your own data

Glossary

- **APK** — Android application package file format
- **Extended Security Updates (ESU)** — Paid/limited continued security patches after official support ends
- **Google Play Protect** — Android's built-in malware-scanning system
- **Microsoft Defender** — Windows' built-in antivirus and firewall
- **Patch** — An update fixing a specific security vulnerability
- **Principle of Least Privilege** — Granting only the minimum necessary access
- **Runtime Permission** — An app's access request made at the moment it's needed
- **Secure Boot** — Ensures only trusted software loads at device startup
- **Sideloading** — Installing an app from outside the official app store
- **TPM (Trusted Platform Module)** — A hardware chip supporting encryption and secure boot

Frequently Asked Questions

1. Why are software updates so important?
They patch known security vulnerabilities that attackers actively target on devices that haven't updated.

2. Is Google Play Protect enough, or do I need separate antivirus?
Play Protect provides strong baseline protection for most users; additional antivirus can add extra features but isn't strictly mandatory for everyone.

3. What is sideloading, and why is it risky?
Installing an app from outside the official app store, bypassing its security review — sideloaded apps were the leading source of new Android malware in 2025.

4. When did Windows 10 support end?
Standard support ended on 14 October 2025.

5. Can I still get security updates for Windows 10?
Yes, through the Extended Security Updates (ESU) program, but only until 13 October 2026 for individual users.

6. What is a runtime permission?
An app's request for access to a specific feature, like camera or location, made at the moment it needs it — which you can grant, deny, or revoke.

7. Should I grant every permission an app asks for?
No — only grant permissions genuinely necessary for the app's core function.

8. What is Microsoft Defender?
Windows' built-in antivirus and firewall protection, enabled by default.

9. Can antivirus protect me if I share my OTP with a scammer?
No — antivirus protects against malicious software, not against voluntarily shared information.

10. What is a security patch?
A software update that specifically fixes a known vulnerability.

11. Is it safe to install an APK sent to me on WhatsApp?
Generally no — install apps through the official Play Store instead, wherever possible.

12. What is TPM, and why does Windows 11 need it?
A hardware security chip supporting encryption and secure boot, raising Windows 11's baseline security requirements.

13. How often should I review my app permissions?
Every few months, or whenever you notice an app behaving unexpectedly.

14. What happens if I keep postponing updates?
Your device remains exposed to vulnerabilities that a patch would have already fixed.

15. What is the Cyber Swachhta Kendra?
A CERT-In initiative providing free malware-removal tools to Indian citizens.

16. Are flashlight or simple utility apps safe to trust with broad permissions?
No — such apps rarely need access to contacts, SMS, or location; treat such requests as a red flag.

17. What's the difference between a feature update and a security patch?
A feature update adds new functionality; a security patch specifically fixes a vulnerability.

18. Should I disable my antivirus to install an app someone told me to over a call?
No — this is a known social-engineering tactic; never disable built-in protections at a stranger's request.

19. Is Windows 11 significantly more secure than Windows 10?
Yes, partly due to mandatory hardware protections like TPM and Secure Boot.

20. What should I do with apps I no longer use?
Uninstall them, rather than leaving them installed with standing permissions.

21. Does having antivirus mean I can skip updates?
No — antivirus and updates address different risks; both remain necessary.

22. Where can I get official help removing malware from my device in India?
Through CERT-In's Cyber Swachhta Kendra initiative.

Suggested Infographics, Flowcharts & Illustrations

(For JanGyan design team — placement suggestions only; no images generated here)

- **"Five Pillars of Device Security" icon diagram** → within "Introduction"

- **Android vs Windows built-in protection comparison card** → within "Fundamental Concepts"
- **Flowchart: "Should I Grant This Permission?"** → within "Complete Detailed Explanation, Part E"
- **Windows 10 support timeline (Oct 2025 → Oct 2026 ESU → beyond)** → within "Complete Detailed Explanation, Part B"
- **"Never Sideload" warning card with APK risk icon** → near "Practical Checklist"

Official References

- Google Security Blog — Android app ecosystem safety report, 2025
- Microsoft — Windows 10 lifecycle and Extended Security Updates (ESU) documentation
- Indian Computer Emergency Response Team (CERT-In) — Cyber Swachhta Kendra and CISG-2025-02 guidelines
- Information Technology Act, 2000 (as amended), India Code portal
- Press Information Bureau (PIB), Government of India — CERT-In annual performance releases

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Official Government Website Links

- CERT-In: **https://www.cert-in.org.in**
- Cyber Swachhta Kendra: **https://www.csk.gov.in**
- MeitY: **https://www.meity.gov.in**
- National Cyber Crime Reporting Portal: **https://cybercrime.gov.in**
- Press Information Bureau: **https://pib.gov.in**

Cybercrime Helpline (24×7, Toll-Free): 1930

Conclusion

Your device's security depends on a handful of small, repeatable habits: keeping it updated, trusting its built-in protections, installing apps only from official stores, and reviewing what permissions you've granted along the way. None of these take much time, and together they close off the vast majority of ways a device gets compromised.

This is only the next step in your Cyber Safety journey with JanGyan. Stay alert, stay informed, and remember: **update promptly, install officially, and grant permissions sparingly.**

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